

Velocity time graphs

- can be used to show uniform and non uniform acceleration of a car traveling along a straight line in one direction

Velocity time graphs: gradient gives acceleration of object

Describing velocity time graphs

1. Direction. + or -, left or right, up or down
2. Acceleration - constant / uniform / increasing / decreasing
When describing, do your best to squeeze in some actual numbers from the graph

When is displacement negative for velocity time graph?

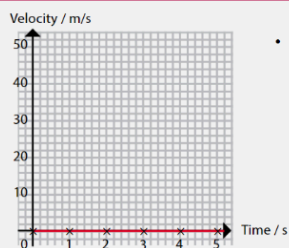
- When the negative region of the graph is greater than the positive region.
- Negative displacement is greater than the positive displacement.

Meaning of negative velocity:

- Object moves in the other direction from which it started.

Car at rest

Time / s	0	1	2	3	4	5
Velocity / m/s	0	0	0	0	0	0

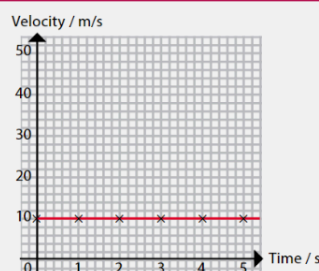


- The velocity of the car remains at 0 m/s so the car has zero acceleration.

- Velocity of car remains at 0, car has 0 acceleration as gradient = 0.
- Area under graph is 0, displacement is constantly 0, stationary

Car travelling at a uniform velocity of 10 m/s

Time / s	0	1	2	3	4	5
Velocity / m/s	10	10	10	10	10	10

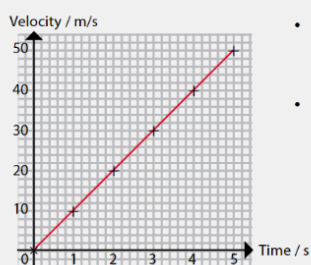


- The velocity of the car remains at 10 m/s so the car has zero acceleration.

- Velocity remains at 10m/s - uniform velocity.
- Car has 0 acceleration as gradient equal to 0.
- Positive direction from starting point [check]

Car travelling with uniform acceleration

Time / s	0	1	2	3	4	5
Velocity / m/s	0	10	20	30	40	50

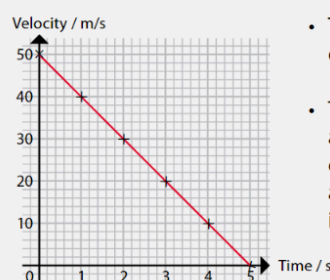


- The velocity of the car increases by 10 m/s every second.
- The graph has a positive and constant gradient, and the acceleration is constant.

- Velocity of car increases by 10 m/s every second
- Graph has positive and constant gradient, acceleration is uniform/constant
- Moves in a positive direction from the starting point.

Car travelling with uniform deceleration

Time / s	0	1	2	3	4	5
Velocity / m/s	50	40	30	20	10	0

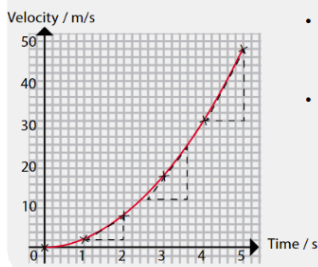


- The velocity of the car decreases by 10 m/s every second.
- The graph has a negative and constant gradient, and the deceleration is constant.

- Velocity of car decreases by 10 m/s every second
- Graph has negative and constant gradient, and there is acceleration in the negative direction (deceleration)
- Object is still moving in the positive direction away from the starting point, but the object has acceleration in the negative direction. (slowing down)

(non-uniform acceleration)

Time / s	0	1	2	3	4	5
Velocity / m/s	0	2	8	18	32	50

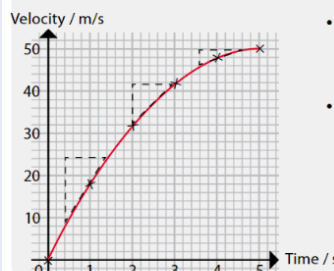


- The increase in velocity of the car is increasing with time.
- The graph has a positive and increasing gradient, and the acceleration increases.

- Increase in velocity of car is increasing with time
- Graph has positive and increasing gradient, acceleration increases
- Moves in a positive direction from the starting point.

Car travelling with decreasing acceleration (non-uniform acceleration)

Time / s	0	1	2	3	4	5
Velocity / m/s	0	18	32	42	48	50



- The increase in velocity of the car is decreasing with time.
- The graph has a positive and decreasing gradient, and the acceleration decreases.

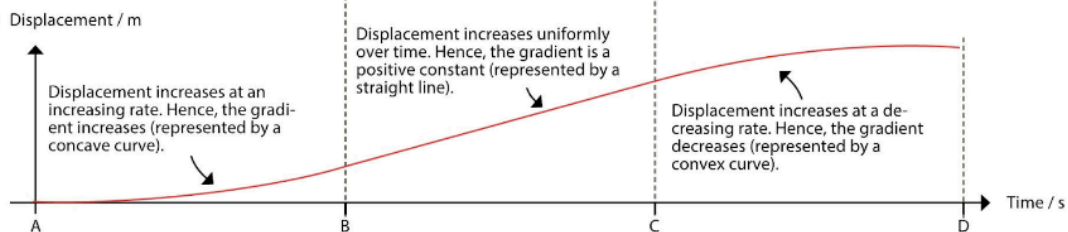
- Increase in velocity of car is decreasing with time
- Graph has a positive and decreasing gradient, and acceleration decreases.
- Moves in a positive direction from the starting point.

Unaccepted phrasing

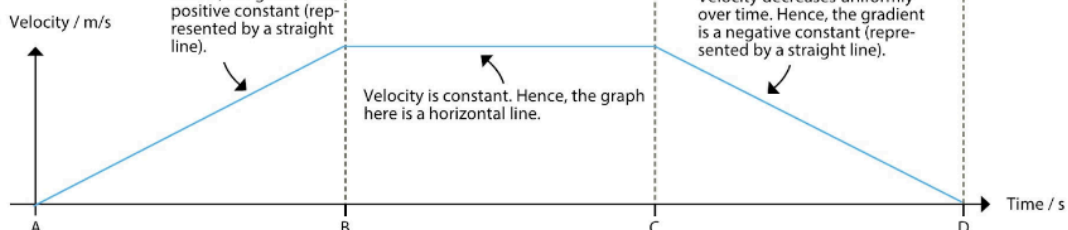
- constantly accelerating
- Rate of acceleration is constant

From A to B	From B to C	From C to D
Car accelerates uniformly from rest.	Car moves at constant velocity.	Car decelerates uniformly to a stop.

The y-axis indicates displacement. Thus, this is a displacement-time graph.



The y-axis indicates velocity. Thus, this is a velocity-time graph.



Area under Velocity Time Graph

⇒ returns displacement

Speed time against graph against velocity time graph

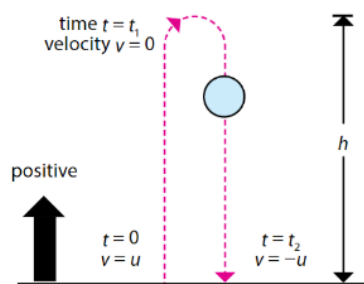


Figure 2.30 Upward and downward journey of a volleyball

assign upward direction from ground as positive

Speed/time graph	Velocity/time graph
$A_1 \rightarrow$ distance traveled h during its upward journey to the highest point. $A_2 \rightarrow$ distance traveled h during its downward journey from the highest point. For speed-time graph: Total distance traveled by volleyball = total area under the speed time graph $A_1 + A_2 = h + h = 2hm$	$A_1 \rightarrow$ displacement is positive during its upward journey to the highest point $A_2 \rightarrow$ displacement is negative during its downward journey from the highest point. For velocity time graph: Total displacement of volleyball = total area under velocity time graph $A_1 + A_2 = h + (-h) = 0m$

Acceleration of free fall

Galileo's finding - all objects, regardless of mass or size, fell at the same acceleration due to earth's gravity

Acceleration due to gravity / gravitational field strength g is a constant.

Value of g is taken to be $10m/s^2$